

## Solution Requirements Specification

|                      |                 |                       |                  |
|----------------------|-----------------|-----------------------|------------------|
| <b>Date:</b>         | 3 October 2026  | <b>Team ID:</b>       | unm1611MADHANRAJ |
| <b>Project Name:</b> | AI FitTrack API | <b>Maximum Marks:</b> | 4 Marks          |

### Functional Requirements (Part 1):

| S.No | FR Category         | Requirement Description                                                                  | Priority | Mapped Endpoint / File                       |
|------|---------------------|------------------------------------------------------------------------------------------|----------|----------------------------------------------|
| 1    | User Registration   | Allow new users to register with name, email, and password. Prevent duplicate emails.    | High     | POST /api/auth/register<br>authController.js |
| 2    | User Authentication | Authenticate user credentials, hash-compare password via bcrypt, issue signed JWT token. | High     | POST /api/auth/login authController.js       |
| 3    | User Profile Access | Retrieve authenticated user's identity and creation date excluding hashed password.      | Medium   | GET /api/auth/profile authController.js      |
| 4    | Create Workout      | Authenticated user can create a workout with name, category, duration, and calories.     | High     | POST /api/workouts<br>workoutController.js   |

### Requirement Overview & Scope:

AI FitTrack API provides complete fitness tracking capabilities coupled with Google Gemini AI intelligence. It ensures strict data isolation so users cannot view or modify workouts belonging to others.

## Functional Requirements (Part 2) & Non-Functional Requirements

|                      |                 |                       |                  |
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### Functional Requirements (Part 2):

| S.No | FR Category           | Requirement Description                                                                 | Priority | Mapped Endpoint / File                              |
|------|-----------------------|-----------------------------------------------------------------------------------------|----------|-----------------------------------------------------|
| 5    | View All Workouts     | Retrieve all workouts logged by authenticated user sorted chronologically (descending). | High     | GET /api/workouts workoutController.js              |
| 6    | View Workout by ID    | Fetch specific workout entry by MongoDB ObjectId with strict user-ownership guard.      | Medium   | GET /api/workouts/:id workoutController.js          |
| 7    | Update Workout        | Allow workout owner to modify duration, category, or calories; return updated document. | Medium   | PUT /api/workouts/:id workoutController.js          |
| 8    | Delete Workout        | Allow workout owner to delete workout entry; enforce 404/403 if unowned.                | Medium   | DELETE /api/workouts/:id workoutController.js       |
| 9    | Multi-Criteria Search | Search workouts via query string (?q=) across workoutName, category, or date.           | High     | GET /api/workouts/search workoutController.js       |
| 10   | AI Recommendation     | Generate personalized workout routines using Gemini AI based on age, goal, experience.  | High     | POST /api/ai/workout-recommendation aiController.js |
| 11   | AI Fitness Insights   | Generate intelligent fitness feedback using Gemini AI based on workout totals.          | High     | POST /api/ai/fitness-insights aiController.js       |

### Non-Functional Requirements (Part 1):

| S.No | NFR Category          | Requirement Description                                                         | Target Metric / Criteria                                     |
|------|-----------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1    | Security & Encryption | Protect passwords and enforce JWT authentication on all workout endpoints.      | Bcrypt 10 rounds; JWT Bearer guard; 0 plaintext credentials. |
| 2    | Latency & Performance | Fast response times for database CRUD operations and external Gen AI calls.     | CRUD < 100ms; Gemini AI < 1.8s response latency.             |
| 3    | Throughput & Scaling  | Support concurrent API requests without thread blocking or dropped connections. | Handle > 150 req/sec under concurrent load.                  |

## Non-Functional Requirements (Part 2) & Traceability Matrix

|                      |                 |                       |                  |
|----------------------|-----------------|-----------------------|------------------|
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### Non-Functional Requirements (Part 2):

| S.No | NFR Category         | Requirement Description                                                          | Target Metric / Criteria                            |
|------|----------------------|----------------------------------------------------------------------------------|-----------------------------------------------------|
| 4    | Reliability & Uptime | Centralized error handling catches all synchronous and asynchronous exceptions.  | Centralized errorHandler; 99.9% operational uptime. |
| 5    | Maintainability      | Decoupled MVC architecture separating routes, controllers, models, and services. | Clean code score > 90%; clear folder layout.        |
| 6    | Portability & Config | Configuration through environment variables (.env); deployable to any cloud.     | Complete .env.example; zero platform lock-in.       |

### Requirements Traceability Matrix (RTM):

| Req ID | Requirement Description | Source Module                      | Verification Test             | Compliance     |
|--------|-------------------------|------------------------------------|-------------------------------|----------------|
| FR-1   | User Registration       | authController.js                  | Postman Auth Suite Test #1    | 100% Compliant |
| FR-2   | User Login & JWT        | authController.js                  | Postman Auth Suite Test #2    | 100% Compliant |
| FR-4   | Add Workout Activity    | workoutController.js               | Postman Workout Suite Test #4 | 100% Compliant |
| FR-5   | View All Workouts       | workoutController.js               | Postman Workout Suite Test #5 | 100% Compliant |
| FR-9   | Workout Regex Search    | workoutController.js               | Postman Search Suite Test #9  | 100% Compliant |
| FR-10  | AI Workout Plan         | aiController.js & geminiService.js | Postman AI Suite Test #10     | 100% Compliant |
| FR-11  | AI Fitness Insights     | aiController.js & geminiService.js | Postman AI Suite Test #11     | 100% Compliant |